



- Index
- Language ▾
- General ▾
- Business ▾
- Science ▾
- Mathematics And Applied Science ▾
- Home Economics ▾
- Civil And Mechanical ▾

Chemistry Paper 3A WASSCE (SC), 2016

- Subject Home
- 1A
- 2A
- 3A
- 1B
- 2B
- 3B

Menu

General
Comments

Weakness/
Remedies

Candidate's
Strength

Question 2A

C is a mixture of two salts, containing one action and two anions.

Carry out the following exercises on C. Record your observations and identify any gas(es) evolved. State the conclusion you draw from the result of each test.

(a) Put all of C in a beaker and add about 10 cm³ of distilled water. Stir well and filter. Keep the filtrate and the residue.

(b) To about 2 cm³ of the filtrate, add few drops of AgNO₃(aq) followed by HNO₃(aq). Add excess NH₃(aq) to the resulting mixture.

(c) (i) Put all of the residue into a clean test tube and add about 5 cm³ of HNO₃(aq).

(ii) To about 2 cm³ of the solution from 2(c)(i), add NaOH(aq) in drops and then in excess.

(iii) To another 2 cm³ of the solution from 2(c)(i), add NH₃(aq) in drops and then in excess.

Observation

Majority of the candidates could not correspond the observations to inferences. In addition, majority of them carried out the tests on a solid sample instead of solutions.

Some candidates lost marks due to wrong charges on ions.

Expected answers include:

Question 2

	TEST	OBSERVATION	INFERENCE
(a)	C _(s) + distilled water + filter	Partly dissolved / soluble Colourless filtrate obtained, white residue	C contains soluble and insoluble salts
(b)	Filtrate of C + AgNO ₃ (aq) + HNO ₃ (aq)	White precipitate Precipitate insoluble	Cl ⁻ / CO ₃ ²⁻ / SO ₃ ²⁻ Cl ⁻

	+ $\text{NH}_3(\text{aq})$	Precipitate soluble/ dissolved	Cl^- present
(c) (i)	Residue + $\text{HNO}_3(\text{aq})$	Effervescence <u>Colourless odourless</u> gas evolved Gas turned lime water milky	CO_2 gas evolved from $\therefore \text{CO}_3^{2-}$ present
(ii)	Solution from c (i) + $\text{NaOH}(\text{aq})$ drops then in excess	White gelatinous precipitate Precipitate dissolved/ soluble	Zn^{2+} / Al^{3+} present Zn^{2+} or Al^{3+} present
(iii)	solution from c (i) + $\text{NH}_3(\text{aq})$ drops Then in excess	White gelatinous precipitate Precipitate dissolved/ soluble	Zn^{2+} / Al^{3+} present (The two ions must be mentioned) Zn^{2+} present

[Prev](#)[Next](#)

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